ANDROID PROGRAMMING PRACTICAL 08

Module 1 · RelativeLayout dynamic alignment

Aim

Use RelativeLayout to align UI components relative to the parent and to other views.

Project name: RelativeLayoutDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create a project and use RelativeLayout as the activity root.

2. Add a title aligned to the parent top/center as needed.

3. Add input fields using rules such as layout_below and alignParentStart.

4. Place buttons relative to each other using layout_toStartOf/layout_toEndOf or equivalent supported attributes.

5. Give referenced views unique IDs and ensure rules refer to existing IDs.

6. Add click listeners for the intended actions.

7. Run the app at different emulator sizes and verify elements align without overlap.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

The UI components align relative to the parent and other views without overlap.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.